

From repeated games to bounded-rational dynamics and learning

Luca Dall'Asta

DISAT
Politecnico di Torino, Italy

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How does dynamics enter in games?

① REPEATED GAMES:

- a normal-form game G is repeatedly played
- players are infinitely rational and far-sighted (maximize discounted cumulative expected utility)
- the equilibrium concepts (Nash and Subgame-Perfect) can be generalized to repeated games

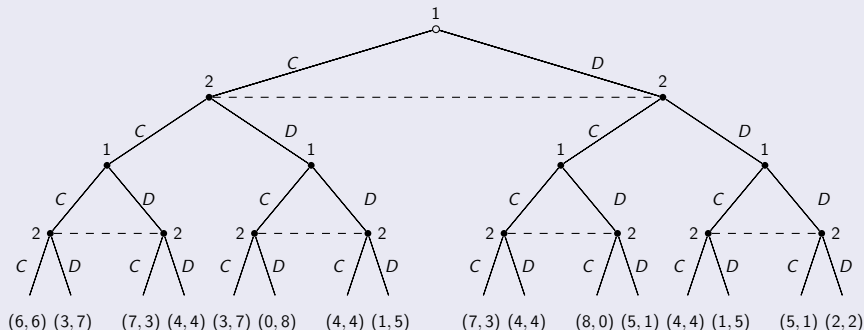
② LEARNING / EQUILIBRIUM SELECTION:

- a normal-form game G is repeatedly played
- players are myopic and follow a behavioral update rule
- the equilibria of G can be dynamically selected
- information about the game can be learned

Finitely Repeated Games: definitions

A **repeated game** is a game in which, at each period/stage of the game, the agents play the same stage game, called *constituent game*, which can be in normal or extensive form. If the number T of repetitions (*horizon*) is finite (and common knowledge), the game is *finitely repeated*.

Example (Repeated Prisoner's Dilemma, $T = 2$)



Finitely Repeated Games: definitions

Two common definitions of utility functions:

- **Average payoff**

$$U_i(h^T) = \frac{1}{T} \sum_{t=1}^T u_i(a^t), \quad \forall i \in \mathcal{N}, \forall h^T \in Z \quad (1)$$

called *Limit of Means* when $T \rightarrow \infty$.

- **Discounted payoff** ($\delta \leq 1$),

$$U_i(h^T) = \sum_{t=1}^T \delta^{t-1} u_i(a^t), \quad \forall i \in \mathcal{N}, \forall h^T \in Z \quad (2)$$

h^T is a terminal history.

Equilibria in Finitely Repeated Games

Definition

A strategy profile $s^* = (s_1^*, \dots, s_N^*)$ is a **Nash equilibrium** of the repeated game Γ if $\forall i \in \mathcal{N}, U_i(s_i^*, s_{-i}^*) \geq U_i(s_i, s_{-i}^*), \forall s_i \in S_i$.

Remark

In a NE of the repeated game, on the equilibrium path, the last period's play must be a Nash equilibrium of the stage game.

In a SPE of the repeated game, this must be valid also off-equilibrium.

Collections/Repetitions of stage-game NEs defines NEs (also SPEs) of the repeated game.

Are there other type of NE (or SPE) of the repeated game?

Equilibria in Finitely Repeated Games

Example

Consider the $T = 2$ repeated game with stage game

	L	C	R
U	7, 1	1, 2	3, 0
M	5, 7	0, 9	3, 4
D	0, 0	0, 1	4, 5

Stage-game NE: (U, C) , (D, R) and $((\frac{2}{3}, 0, \frac{1}{3}), (0, \frac{1}{2}, \frac{1}{2}))$ with expected payoff $(2, \frac{5}{3})$.

(U, C) is also the stage-game minmax strategy profile, indeed

$$\bar{v}_1 = \min_{a_2 \in A_2} \max_{a_1 \in A_1} u_1(a_1, a_2) = \min_{a_2 \in A_2} \{7, 1, 4\} = 1,$$

$$\bar{v}_2 = \min_{a_1 \in A_1} \max_{a_2 \in A_2} u_2(a_1, a_2) = \min_{a_1 \in A_1} \{2, 9, 5\} = 2.$$

Equilibria in Finitely Repeated Games

Example (cont'd)

At $t = 2$, player 1 can use the stage-game NE (U, C) (resp. (D, R)) to punish (resp. reward) player 2.

Consider the strategy profile $s = (s_1, s_2)$ with

$$s^1(h^0) = (M, L)$$

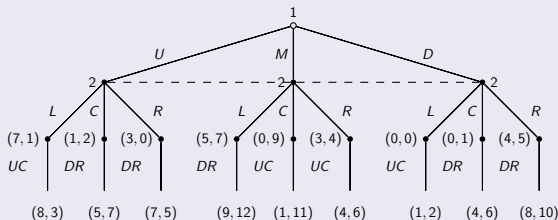
$$s^2(h^1) = \begin{cases} (D, R) & \text{if } h^1 = (M, L) \text{ or if } h^1 = (U/D, C/R) \\ (U, C) & \text{if } h^1 = (U/D, L) \text{ or } h^1 = (M, C/R). \end{cases}$$

The existence of this SPE is based on the **credible threat** of punishment.

Equilibria in Finitely Repeated Games

Example (cont'd)

Applying backward induction,



	$t = 1$ conditional game		
	L	C	R
U	8, 3	5, 7	7, 5
M	<u>9, 12</u>	1, 11	4, 6
D	1, 2	4, 6	8, 10

Equilibria in Finitely Repeated Games: general case

If a NE of the stage game exceeds the minmax payoff, the minmax strategy can be used to punish a player for deviating.

Definition

A (strictly) **enforceable payoff profile** is a payoff profile w for which $w_i \geq \bar{v}_i \forall i \in \mathcal{N}$ (resp. $w_i > \bar{v}_i, \forall i \in \mathcal{N}$). If $u(a)$ is (strictly) enforceable, we call the profile a a (strictly) enforceable outcome of G .

Theorem (NASH-FOLK TH. FINITELY REPEATED GAMES)

If the constituent game $G = (\mathcal{N}, A, u)$ admits a (unique) Nash equilibrium in pure strategies $\hat{a}$ in which the payoff of every player i exceeds his minmax payoff $\bar{v}_i$, then for any strictly enforceable outcome a^ of G and any $\epsilon > 0$, there exists an integer T^* such that, if $T > T^*$, the repeated game of G with horizon T has a Nash equilibrium in which the payoff of each player i is within ϵ of $u_i(a^*)$.*

Equilibria in Finitely Repeated Games: general case

Proof.

$$s^t = \begin{cases} s^{*,t} = (a_1^*, \dots, a_N^*) & \forall t \leq T - L, \\ \hat{s}^t = (\hat{a}_1, \dots, \hat{a}_N) & \forall t > T - L \\ p_{-j} = \arg \min_{a_{-j} \in A_{-j}} \max_{a_j \in A_j} u_j(a_j, a_{-j}) & \forall t > t', \\ \quad \text{if } \exists! j \in \mathcal{N} \text{ deviated from } s^{*,t'} \text{ for } t' \leq T - L, \end{cases}$$

A deviation for $t \leq T - L$ is deterred if L such that

$$\max_{a_j \in A_j} u_j(a_j, a_{-j}^*) - u_j(a^*) \leq L(u_j(\hat{a}) - \bar{v}_j), \quad \forall j \in \mathcal{N}.$$

To obtain a payoff profile within ϵ of $u(a^*)$, choose T^* s.t. $\forall T > T^*$

$$|T^{-1} \{(T - L) u_i(a^*) + L u_i(\hat{a})\} - u_i(a^*)| < \epsilon, \quad \forall i \in \mathcal{N}.$$



Infinitely Repeated Games

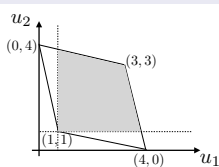
If the game is repeated an infinite number of times (or T is not known). What average payoff can be sustained?

Definition

A **feasible payoff profile** $v \in \mathbb{R}^N$ of $G = (\mathcal{N}, A, u)$ is a convex combination of payoff profiles of outcomes in A , i.e. $v = \sum_{a \in A} \alpha_a u(a)$ with $\alpha_a \geq 0$, $\forall a \in A$ and $\sum_{a \in A} \alpha_a = 1$.

Example

In the PD, a feasible enforceable payoff profiles requires $u_i \geq 1$ for $i = 1, 2$.



Infinitely Repeated Games: Folks theorems

Can we obtain feasible enforceable payoff profiles in infinitely repeated games using simplified types of strategies?

Automaton strategies: players have states and play a strategy associated to that state until a change in the behavior of other players make them switch to another state.

Example

Grim-trigger strategy in the Prisoner's Dilemma: play C as long as the others do, switch to D forever to punish any deviation to D .

Theorem (NASH-FOLK TH. FOR DISCOUNTING CRIT.)

Let w be a strictly enforceable feasible payoff profile of $G = (\mathcal{N}, A, u)$, $\forall \epsilon > 0$, $\exists \underline{\delta} < 1$ such that, if $\delta > \underline{\delta}$ then the δ -discounted infinitely repeated game of G has a N.E. whose payoff profile w' satisfies $|w - w'| < \epsilon$.

Infinitely Repeated Games: Folks theorems

Example (COOPERATION IN THE PRISONER'S DILEMMA)

Consider the symmetric Grim-Trigger strategy profile. Then

- ① if D was played by one player, the other will also play D and both continue forever, receiving the subgame-perfect payoff for the remaining times.
- ② If D was not yet played, then players have the alternatives:
 - play C , with remaining payoff

$$\Delta U^> = (1 - \delta) (3 + 3\delta + 3\delta^2 + \dots) = 3$$

- play D with

$$\Delta U^> = (1 - \delta) (4 + \delta + \delta^2 + \dots) = 4(1 - \delta) + \delta$$

Cooperating is profitable if $3 \geq 4(1 - \delta) + \delta$, i.e. if $\delta \geq 1/3$.

The Grim-Trigger strategy profile is a Nash equilibrium for $\delta \geq 1/3$.

Infinitely Repeated Games: Folks theorems

Example (TIT-FOR-TAT)

Tit-for-Tat strategy: initially play C , then play the action chosen by the other in the previous period.

If both players use Tit-for-Tat, they get payoff 3 at each round.

Suppose that player 2 deviates at t and comes back to cooperation after $k \geq 1$ rounds. The payoff difference obtained deviating is

$$\Delta U_2^> = (1 - \delta) \left(4 + \delta + \delta^2 + \dots + \delta^k \right) = 4(1 - \delta) + \delta - \delta^{k+1}, \quad (3)$$

to be compared with $3(1 - \delta^{k+1})$ from cooperating:

$$3(1 - \delta^{k+1}) \geq 4(1 - \delta) + \delta - \delta^{k+1} \implies 3\delta - 2\delta^{k+1} - 1 \geq 0, \quad (4)$$

In general, $\forall k \geq 1, \exists \delta_k^*$ s.t. $\forall \delta > \delta_k^*$ the Tit-for-Tat strategy profile is a Nash equilibrium.

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Dynamical systems and stability

Consider a discrete dynamical system $x^{t+1} = f_t(x^t)$ for $t \in [0, \infty)$.

Given x^0 , the *flow map* $x^t = F_t(x^0)$ is such that $F_{t+1}(x^0) = f_t(F_t(x^0))$.

Definition

A state x^* such that $F_t(x^*) = x^*$, $\forall t > 0$ is a **fixed point** of flow F .

A fixed point x^* is **stable** if for every neighborhood $U(x^*)$, then $\exists U'(x^*)$ with $U'(x^*) \subseteq U(x^*)$ such that if $x^0 \in U'(x^*)$, then $F_t(x^0) \in U(x^*)$, $\forall t > 0$.

A fixed point x^* is **asymptotically stable** if it is stable and $\forall x^0 \in U'(x^*)$ then $\lim_{t \rightarrow \infty} F_t(x^0) = x^*$.

The basin of attraction of an asymptotically stable f.p. is the set $\{x^0 \mid \lim_{t \rightarrow \infty} F_t(x^0) = x^*\}$.

A fixed point is **globally stable** if unique and its basin of attraction equals the whole state space.

Best Response dynamics

In the **Best Response Dynamics** (BRD), given an initial profile a^0 , at each time t , a revision of individual actions takes place according to the (synchronous/sequential) adjustment rule

$$a_i^{t+1} \in B_i(a_{-i}^t) = \arg \max_{a_i} u_i(a_i, a_{-i}^t).$$

Proposition

An action profile is a fixed point of Best Response dynamics if and only if it is a pure-strategy NE.

Example

In the Cournot model of competition, the BRD converges to the unique (pure-strategy) Nash equilibrium from any initial conditions (globally stable).

Best Response dynamics: convergence

Remark

We say that Best Response dynamics **converges** if the process ends independently of the initial conditions and the update order.

Convergence to a N.E. is not guaranteed in general games.

Example

	A	B	C
A	$1, -1$	$-1, 1$	$-2, -2$
B	$-1, 1$	$1, -1$	$-2, -2$
C	$-2, -2$	$-2, -2$	$2, 2$

The psNE is unique but it is not globally stable. There is a cycle due to the existence of a mixed NE.

Best Response dynamics: convergence

Definition

A function $V : A \rightarrow \mathbb{Z}$ is a **weakly ordinal potential** function for $G = (\mathcal{N}, A, u)$ if $\forall a = (a_i, a_{-i}) \in A$, $\forall i \in \mathcal{N}$, and $\forall a'_i \in A_i$ such that $a_i \notin B_i(a_{-i})$ but $a'_i \in B_i(a_{-i})$, then $V(a'_i, a_{-i}) > V(a_i, a_{-i})$.

Example

	l	r
T	2, 2	0, 3
B	3, 0	1, 2

(B, r) unique psNE. Set $V(B, r) = 0$, then count payoff improvements

$$V(B, l) = V(B, r) - 1 = -1$$

$$V(T, r) = V(B, r) - 1 = -1$$

$$V(T, l) = V(T, r) - 1 = V(B, l) - 1 = -2$$

Best Response dynamics: convergence

Proposition

BRD converges in G if and only if G admits a weakly ordinal potential.

Proof.

($\Leftarrow$) V increases each time a player best-responds and the game is finite, so a local maximum of potential is reached in a finite number of steps.

($\Rightarrow$) Draw the best-response graph of G (nodes are action profiles and directed edges are profitable deviations) and define $V(a) = -\ell(a)$ where $\ell(a)$ is the length of longest path from a to any sink of the graph (a psNE). Under BRD, ℓ always decreases, so V increases. □

Bounded Rationality

The bounded rationality of agents can be described by a utility model with **random perturbations** $\{\epsilon_i\}$

$$U_i^\epsilon(a) = u_i(a_i, a_{-i}) + \epsilon_i(a_i)$$

The corresponding best response dynamics becomes

$$a_i^{t+1} \in B_i^\epsilon(a_{-i}^t) = \arg \max_{a_i} \{u_i(a_i, a_{-i}^t) + \epsilon_i^t(a_i)\}.$$

The dynamics defines a Markov chain that admits an invariant measure.

In the limit of vanishing noise ($\epsilon_i^t \rightarrow 0$), the invariant measure concentrates on a subset of action profiles called **stochastically stable states** (SSS).

Bounded Rationality: Smoothed Best Response

A special case is obtained when $\epsilon_i^t(a_i) = \epsilon_i^t/\beta$ and $\epsilon_i^t \sim \text{Gumbel}(0, 1)$ (extreme value theory).

The corresponding **Smoothed (Quantal) Best Response** (SBRD) is equivalent to a Markov chain with transition probability

$$\Pr[a_i^{t+1} = a_i | a_{-i}^t] = \frac{e^{\beta u_i(a_i, a_{-i}^t)}}{\sum_{a'_i} e^{\beta u_i(a'_i, a_{-i}^t)}}$$

with inverse temperature $\beta \geq 0$.

Definition

A normal-form game $G = (\mathcal{N}, A, u)$ is a **potential game** if it exists a function $V : A \rightarrow \mathbb{R}$ such that $\forall i \in \mathcal{N}, \forall a_{-i} \in A_{-i}$ and $a_i, a'_i \in A_i$,

$$u_i(a_i, a_{-i}) - u_i(a'_i, a_{-i}) = V(a_i, a_{-i}) - V(a'_i, a_{-i}). \quad (5)$$

The (pure-strategy) Nash equilibrium condition corresponds to maximizing (locally) the potential function $V(a)$.

Theorem (Monderer & Shapley '96)

Every finite potential game has a pure-strategy Nash equilibrium.

Potential games

Example

Prisoner's Dilemma		<i>C</i>	<i>D</i>	admits potential		<i>C</i>	<i>D</i>
	<i>C</i>	2, 2	0, 3		<i>C</i>	0	1
	<i>D</i>	3, 0	1, 1		<i>D</i>	1	2

Battle of Sexes		<i>A</i>	<i>C</i>	admits potential		<i>A</i>	<i>C</i>
	<i>A</i>	2, 1	0, 0		<i>A</i>	2	1
	<i>C</i>	0, 0	1, 2		<i>C</i>	0	2

The n -firm Cournot model $u_i(q_i, q_{-i}) = q_i \left(a - \sum_j q_j \right) - cq_i$ admits potential

$$V(q) = (a - c) \sum_{i=1}^n q_i - \sum_{i=1}^n q_i^2 - \sum_{i \neq j} q_i q_j.$$

Convergence in potential games

Both BRD and SBRD converge in potential games.

Proposition

In any finite potential game, best response dynamics always converge to a pure-strategy Nash equilibrium.

Indeed, by definition, under best response the potential cannot decrease (no cycles are possible) and **psNE are local maxima** of the potential.

Proposition

*In potential games, **stochastically stable states** coincide with the set of pure-strategy Nash equilibria corresponding to the **global maximum** of the potential function.*

Convergence in potential games

Example (Minimum Effort Game)

EU countries want to guarantee borders protection. Increasing levels of security $a_i \in \{1, 2, \dots, K\}$. Overall security level is set by the country with lower security standards

$$u_i(a) = b \min \{a_1, a_2, \dots, a_n\} - ca_i.$$

There are K symmetric psNE with $a_i = a^*, \forall i$. The most efficient (Pareto dominant) is given by $a^* = K$. The game admits a potential

$$V(a) = b \min \{a_1, a_2, \dots, a_n\} - c \sum_i a_i.$$

Maximization of the potential gives $V_{\max} = (b - nc) a^*$ with $a_i = a^*, \forall i$. For $N > b/c$, the potential is maximized by the **less efficient NE!**

Replicator Equations

What about dynamics in **mixed strategies**?

In a mixed-strategy equilibrium:

- actions in the support of the mixed strategy give the same expected utility,
- actions outside the support have a lower expected utility.

Consider an internal process in player i 's mind such that:

- successful strategies with higher payoffs grow
- less successful ones decay.

Remark

In **Evolutionary Game Theory**, replicator equations represent the “natural” evolutionary dynamics of strategies in a population of agents (without mutation).

Replicator Equations

Let i play a_i at time t with probability $\pi_i^t(a_i)$ and define the dynamics

$$\pi_i^{t+1}(a_i) = \pi_i^t(a_i) + \alpha \pi_i^t(a_i) \left[\bar{u}_i^t(a_i) - \sum_{a'_i \in A_i} \pi_i^t(a'_i) \bar{u}_i^t(a'_i) \right]$$

with $\sum_a \pi_i^t(a) = 1, \forall t \geq 0$ and

$$\bar{u}_i^t(a_i) = \sum_{a_{-i} \in A_{-i}} u_i(a_i, a_{-i}) \prod_{j \in \mathcal{N} \setminus i} \pi_j^t(a_j).$$

In continuous time, we obtain the **Replicator Equations**

$$\frac{d}{dt} \pi_i^t(a_i) = \pi_i^t(a_i) \left[\bar{u}_i^t(a_i) - \sum_{a'_i \in A_i} \pi_i^t(a'_i) \bar{u}_i^t(a'_i) \right]$$

Replicator Equations

Most results on convergence were obtained in EGT, so they hold only for **symmetric games**. Not guaranteed for general games.

Proposition

If the strategy profile σ^ is a mixed-strategy NE of a finite normal-form game G , then the vector $\pi = \sigma^*$ is a fixed point of the replicator equations for G .*

However, not every fixed point is a NE of the game.

Proposition

Given a finite normal-form game G , if the vector π^ is a **stable** fixed point of the replicator equations for G , then the strategy profile $\sigma = \pi^*$ is a mixed-strategy NE of G .*

Replicator Equations

Example

Consider again the Hawk-Dove game

	H	D
H	0, 0	5, 1
D	1, 5	3, 3

$$\begin{aligned}\frac{d}{dt}\pi^t(H) &= \pi^t(H) \left[\bar{u}^t(H) - \sum_{a' \in A} \pi^t(a') \bar{u}^t(a') \right] \\ &= \pi^t(H) (1 - \pi^t(H)) (2 - 3\pi^t(H))\end{aligned}$$

The strategy $\sigma = (2/3, 1/3)$ is the only asymptotically stable state of Replicator Equations.

Fictitious Play

Learning rule: player i forms beliefs $\{\mu_i^t\}_{t \geq 0}$ about opponents' play and behave rationally with respect to them.

Let $\eta_i^t : A_{-i} \rightarrow \mathbb{N}$ be the number of times i has observed the partial profile a_{-i} up to time $t - 1$, with arbitrary η_i^0 and define the belief of player i as the **empirical frequency distribution**

$$\mu_i^t(a_{-i}) = \frac{\eta_i^t(a_{-i})}{\sum_{a_{-i} \in A_{-i}} \eta_i^t(a_{-i})}.$$

Then, a_i^t is chosen to be

$$a_i^t \in \arg \max_{a_i} u_i(a_i, \mu_i^t).$$

Remark

Players do not have to know anything about opponents' payoffs, only about how their play.

Fictitious Play

In two-player games, players in fictitious play assume that the opponent is using a *stationary mixed strategy*.

Player i believes that play by $-i$ is a sequence of i.i.d. multinomial random variables with a fixed but unknown distribution $\sigma_{-i}(a_{-i})$.

Given a realization κ_i of $K = t' - t$ outcomes with $K = \sum_{a_{-i}} \kappa_i(a_{-i})$,

$$\Pr[\kappa_i | \sigma_{-i}] = \frac{K!}{\prod_{a_{-i}} \kappa_i(a_{-i})!} \prod_{a_{-i}} \sigma_{-i}(a_{-i})^{\kappa_i(a_{-i})}$$

Player i 's prior belief over $\sigma_{-i}(a_{-i})$ at time t takes the form of a *Dirichlet distribution*

$$\Pr[\sigma_{-i}] = \nu_i^t(\sigma_{-i}) = \frac{\Gamma\left(\sum_{a_{-i}} \eta_i^t(a_{-i})\right)}{\prod_{a_{-i}} \Gamma(\eta_i^t(a_{-i}))} \prod_{a_{-i}} \sigma_{-i}(a_{-i})^{\eta_i^t(a_{-i})-1}.$$

Fictitious Play

The posterior distribution is given by Bayesian inference

$$\begin{aligned}\Pr[\sigma_{-i}|\kappa_i] &= \frac{\Pr[\kappa_i|\sigma_{-i}] \nu_i^t(\sigma_{-i})}{\int d\sigma_{-i} \Pr[\kappa_i|\sigma_{-i}] \nu_i^t(\sigma_{-i})} \\ &\propto \prod_{a_{-i}} \sigma_{-i}(a_{-i})^{\eta_i^t(a_{-i}) + \kappa_i(a_{-i}) - 1},\end{aligned}$$

that implies the Bayesian updating

$$\begin{aligned}\mu_i^{t'}(a_{-i}) &= \int_{\Sigma_{-i}} \sigma_{-i}(a_{-i}) \Pr[\sigma_{-i}|\kappa_i] d\sigma_{-i} \\ &= \frac{\eta_i^t(a_{-i}) + \kappa_i(a_{-i})}{\sum_{a_{-i}} \eta_i^t(a_{-i}) + \kappa_i(a_{-i})} = \frac{\eta_i^{t'}(a_{-i})}{\sum_{a_{-i}} \eta_i^{t'}(a_{-i})}\end{aligned}$$

Remark

Even if the update is Bayesian, the prior belief, that players play stationary strategies, might be wrong.

Fictitious Play

Proposition

If the sequence $\{a^t\}_{t \geq 0}$ generated by fictitious play for a normal-form game G is such that $\forall i, \forall a_i \in A_i$

$$\lim_{T \rightarrow \infty} \frac{1}{T+1} \sum_{t \geq 0} \mathbb{I}[a_i^t = a_i] = \sigma_i(a_i),$$

where σ is a mixed strategy profile, then σ is a Nash equilibrium of G .

Proposition

Given the sequence $\{a^t\}_{t \geq 0}$ generated by fictitious play applied to G :

- ① *if $\exists \hat{t}$ such that $a^{\hat{t}} = a^*$, where a^* is a strict Nash equilibrium of G , then $a^t = a^* \forall t > \hat{t}$;*
- ② *if $\exists \hat{t}$ such that $a^t = a^* \forall t \geq \hat{t}$, then a^* is a (possibly weak) Nash equilibrium of G .*

Fictitious Play: issues

- Convergence to a mixed strategy eq is guaranteed for few classes of games:
 - zero-sum games,
 - 2×2 games,
 - games solvable by IESDS,
 - potential games.

However, convergence in time average does not imply the N.E. is played, i.e. payoffs can be lower.

- Better convergence can be achieved using smoothed and moderated versions.
- Multi-player fictitious play implies an ambiguous definition of prior belief.

Reinforcement Learning

Consider a learning dynamics in which, at time t , each player i has

- a belief $\mu_i^t(a_{-i})$ about opponents' strategies (e.g. updated like in fictitious play),
- a Q-function $Q_i^t(a)$ for every profile $a \in A$, updated by

$$Q_i^{t+1}(a_i, a_{-i}) = (1 - \alpha) Q_i^t(a_i, a_{-i}) + \alpha \left\{ u_i^t(a_i, a_{-i}) + \delta \max_{a'_i \in A_i} \sum_{a'_{-i} \in A_{-i}} Q_i^t(a'_i, a'_{-i}) \mu_i^t(a'_{-i}) \right\}$$

and let agent i play action a_i at time t with probability

$$\pi_i^t(a_i) = \frac{e^{\beta \tilde{Q}_i^t(a_i)}}{\sum_{a'_i \in A_i} e^{\beta \tilde{Q}_i^t(a'_i)}}$$

where $\tilde{Q}_i^t(a_i) = \sum_{a_{-i} \in A_{-i}} Q_i^t(a_i, a_{-i}) \mu_i^t(a_{-i})$.

- General principle: RL dynamics is based on *exploitation/exploration tradeoff*
- Convergence depends on the properties of the game and on parameters (α, β)
- the continuous-time limit (Sato-Crutchfield equations) is more suitable for analytical study (finite β replicator dynamics)
- RL can be generalized to stochastic games (bounded-rational agents interact in a stochastic environment).